

Expression Tree Construction Report

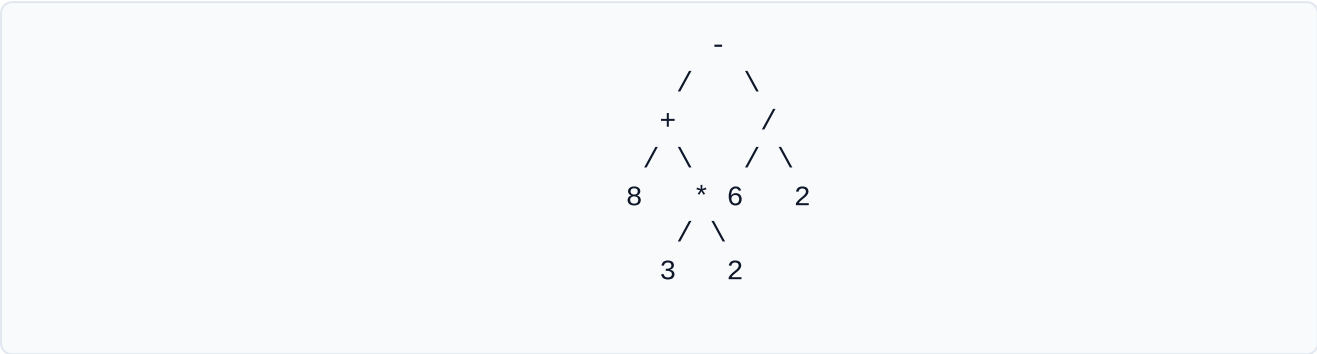
Data Structures & Algorithms Execution Report - Part 1

Target Postfix Expression: 8 3 2 * + 6 2 / -
Final Root: -

ALGORITHM EXECUTION TRACE

Token	Operation & Stack State
8	Create node 8 and push to stack.
3	Create node 3 and push to stack.
2	Create node 2 and push to stack.
*	Pop 2 and 3, create * node with 3 (left) and 2 (right); push result.
+	Pop * and 8, create + node with 8 (left) and * (right); push result.
6	Create node 6 and push to stack.
2	Create node 2 and push to stack.
/	Pop 2 and 6, create / node with 6 (left) and 2 (right); push result.
-	Pop / and +, create - node with + (left) and / (right); push final root.

RESULTING EXPRESSION TREE



Postfix Evaluation Trace

Data Structures & Algorithms Execution Report - Part 2

Expression: 8 3 2 * + 6 2 / -
Final Result: 11

STEP-BY-STEP EVALUATION TRACE

Token	Operation	Stack State
8	Push 8	8
3	Push 3	8, 3
2	Push 2	8, 3, 2
*	$3 * 2 = 6$	8, 6
+	$8 + 6 = 14$	14
6	Push 6	14, 6
2	Push 2	14, 6, 2
/	$6 / 2 = 3$	14, 3
-	$14 - 3 = 11$	11

Final Computed Result: 11